

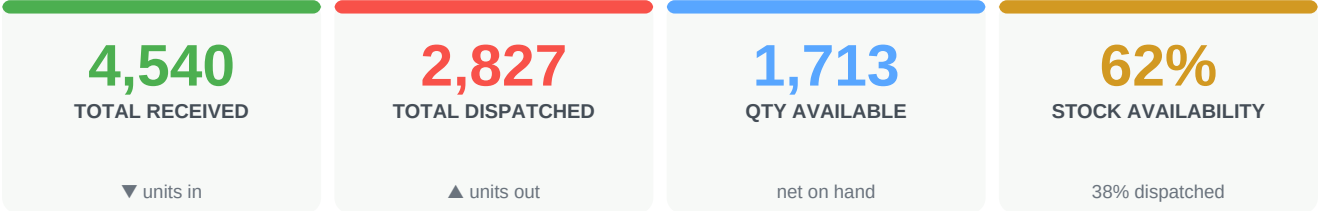
INV

Inventory Management Dashboard

Excel · PivotTables · Data Analysis · Business Intelligence

A production-ready, zero-macro inventory system demonstrating advanced Excel engineering

20 Products5 Shelves62% Available

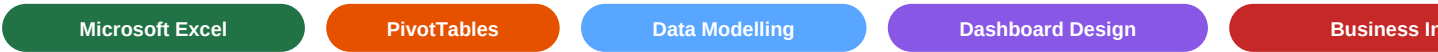


Key performance indicators — live figures from the dashboard workbook

01 · ABOUT THE PROJECT

This project delivers a **fully interactive, zero-macro inventory management system** built entirely in Microsoft Excel. It tracks stock movements — receipts and dispatches — for 20 products across 5 warehouse shelves, calculates real-time availability, and surfaces the lowest-stock items to prevent stockouts. The dashboard updates automatically on every slicer or filter interaction, with no VBA, no plugins, and no internet connection required.

The system is designed as both a **production-ready business tool** and a **data analysis portfolio piece**, demonstrating advanced spreadsheet engineering, structured data modelling, PivotTable-powered calculations, and dashboard UX design principles.



02 · DASHBOARD OVERVIEW

62%

available

The dashboard sheet provides a single-screen view of warehouse health:

- **4 Live KPI Cards** — Total Received, Total Dispatched, Qty Available, and % Stock Availability
- **12-Month Trend Chart** — area line chart showing stock availability from January to December
- **Bottom 10 Table** — colour-coded rows (Critical / Low / OK) to prioritise reorder decisions
- **Interactive Slicers** — Shelf and Product filters that update all visuals simultaneously
- **Reorder Intelligence** — per-product Re-Order Level and Re-Order Quantity thresholds

Fig 1. Donut chart showing 62% stock availability (green = available, grey = dispatched)

03 · DATA VISUALISATIONS

Stock Availability Trend — Jan to Dec

Monthly available units across all products

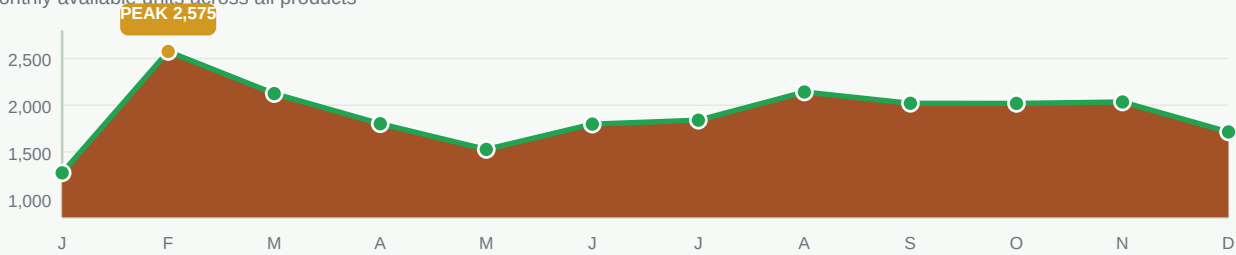


Fig 2. Monthly stock availability trend — peak in February (2,575 units), lowest in January (1,275 units)

Monthly Breakdown

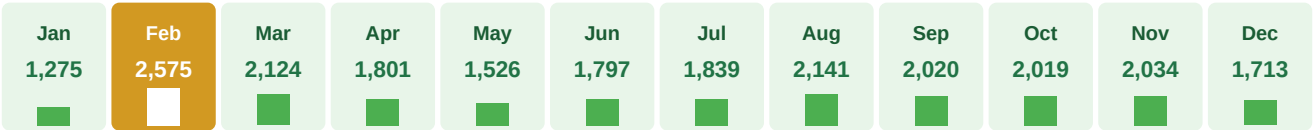


Table 1. Units available per month — amber cell highlights the annual peak (February, 2,575 units)

Bottom 10 Stock Quantity by Product

Products ranked by current quantity available

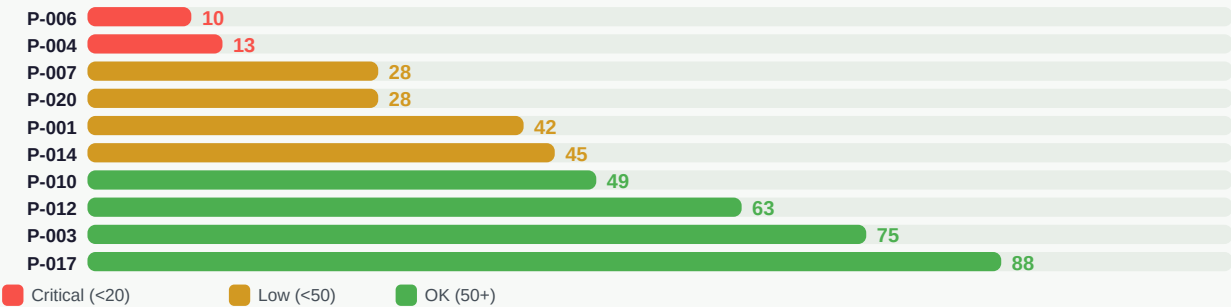


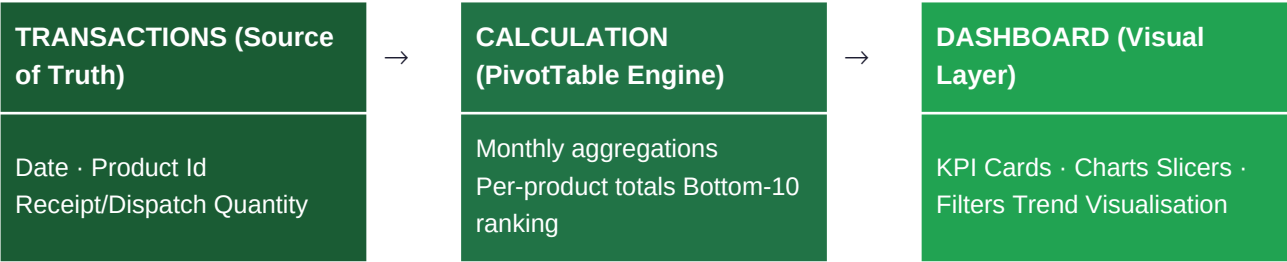
Fig 3. Bottom 10 products by stock quantity — red bars indicate critical reorder levels (<20 units)

04 · DATA ARCHITECTURE & WORKBOOK STRUCTURE

Sheet	Role	Key Columns	Records
■ Dashboard	Front-end visual layer — KPI cards, trend chart, bottom-10 table, slicers	Linked to Calculation via PivotTable	—
■ Products	Master product catalogue — single source of truth for product metadata	Product Id, Name, Shelf, Re-Order Level, Re-Order Qty	20 rows
■ Transactions	Immutable event log — all stock movements recorded chronologically	Date, Product Id, Transaction Type, Quantity	100+ rows
■ ■ Calculation	PivotTable engine — aggregates raw transactions into dashboard-ready summaries	Monthly totals, per-product qty, bottom-10 ranking, availability %	Dynamic

Table 2. Four-sheet workbook architecture — normalised data model separating raw data, processing, and presentation layers

Data Flow Pipeline



05 · PRODUCT CATALOGUE & STOCK STATUS

ID	Product	Shelf	Reorder Lvl	Reorder Qty	Qty Avail.	Status
P-001	Product 1	Shelf 1	48	120	42	LOW
P-002	Product 2	Shelf 2	46	115	130	OK
P-003	Product 3	Shelf 3	60	150	75	OK
P-004	Product 4	Shelf 4	32	110	13	CRITICAL
P-005	Product 5	Shelf 5	30	110	98	OK
P-006	Product 6	Shelf 1	40	100	10	CRITICAL
P-007	Product 7	Shelf 2	46	110	28	LOW
P-008	Product 8	Shelf 3	56	120	166	OK
P-009	Product 9	Shelf 4	70	150	104	OK
P-010	Product 10	Shelf 5	28	100	49	LOW
P-011	Product 11	Shelf 1	40	100	128	OK
P-012	Product 12	Shelf 2	52	120	63	OK
P-013	Product 13	Shelf 3	34	105	132	OK
P-014	Product 14	Shelf 4	40	105	45	LOW
P-015	Product 15	Shelf 5	38	105	130	OK
P-016	Product 16	Shelf 1	36	105	125	OK
P-017	Product 17	Shelf 2	54	115	88	OK
P-018	Product 18	Shelf 3	50	115	134	OK
P-019	Product 19	Shelf 4	34	100	125	OK
P-020	Product 20	Shelf 5	32	100	28	LOW

Table 3. Complete product catalogue with current stock levels and reorder parameters. Red rows = critical (<20 units), amber = low stock.

06 · KEY INSIGHTS & BUSINESS FINDINGS

■ Critical Reorder Alert

P-006 (Product 6, Shelf 1) has only 10 units remaining — 75% below its reorder level of 40. P-004 (Product 4, Shelf 4) has 13 units against a reorder level of 32. Both require immediate replenishment orders.

■ 5 Products Below Reorder

Five products (P-001, P-007, P-014, P-020, P-010) are below their configured reorder levels. These should be included in the next purchasing cycle to prevent stockouts within the forecast period.

■ February Stock Peak	February saw the highest stock availability at 2,575 units — likely reflecting a large seasonal receipt cycle. January was the lowest at 1,275 units, suggesting a post-holiday draw-down pattern in the supply chain.
■ 62% Overall Availability	Of the 4,540 total units received, 1,713 remain available (37.7% dispatched). The 62% availability rate indicates healthy overall stock health, though the critical items skew this figure and need attention.
■ Best Stocked Products	Product 8 (166 units, Shelf 3) and Product 18 (134 units, Shelf 3) are the best stocked items, both significantly above their reorder levels. Shelf 3 has the strongest overall stock position across its product lines.

07 · SKILLS DEMONSTRATED

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	Multi-sheet relational structure, foreign keys via Product Id	Products ↔ Transactions ↔ Calculation	★★★★★
PivotTable Engineering	GETPIVOTDATA, dynamic ranges, multi-field aggregation	Calculation sheet → Dashboard	★★★★★
	Line chart, area fill, bar chart, donut chart, trend analysis	Dashboard trend & bottom-10 charts	★★★★★
Dashboard UX Design	KPI card layout, colour-coded alerts, consistent visual hierarchy	Dashboard sheet	★★★★■

	Cross-sheet interactive filtering, PivotTable connections	Shelf and Product slicers	★★★★■
Excel Formulas	SUMIF, COUNTIF, dynamic named ranges, conditional logic	Calculation & Products sheets	★★★★■
	KPI definition, reorder logic, stock health metrics, trend reading	Full project scope	★★★★■

08 · HOW TO USE THIS WORKBOOK

Step 1	Download & Open Download the .xlsx file from GitHub. Open in Microsoft Excel 2016 or later. Click "Enable Editing" if prompted — no macros will run.
Step 2	Explore Dashboard Navigate to the Dashboard sheet. All KPIs and charts are live. Use the Shelf and Product Name slicers at the top to filter all visuals simultaneously.
Step 3	Log Transactions Go to the Transactions sheet. Add rows with: Date, Product Id (e.g. P-001), Transaction Type (Receipt/Dispatch), and Quantity (positive for receipts, negative for dispatches).
Step 4	Add Products Go to the Products sheet. Add a new row with a unique Product Id (P-021+), name, shelf assignment, Re-Order Level, and Re-Order Quantity.
Step 5	Refresh Dashboard After adding new data, go to Data → Refresh All. All PivotTables, KPIs, and charts update automatically to reflect the new data.

09 · REQUIREMENTS & COMPATIBILITY

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Microsoft Excel	2016 or later	Required for PivotTable slicers and dynamic charts
Operating System	Windows 10/11 or macOS 10.14+	Fully tested on both platforms
Macros / VBA	Not required	Zero code — works with standard Excel features only
Internet Access	Not required	Fully offline — no external data connections
File Size	~45 KB	Lightweight — fast to open and share via email or GitHub
Dependencies	None	No add-ins, plugins, or external libraries needed

10 · PROJECT LINKS & CONTACT

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GitHub Repository	github.com/YOUR-USERNAME/inventory-management-dashboard	Source code, README, and downloadable .xlsx file
LinkedIn Profile	linkedin.com/in/YOUR-PROFILE	Professional profile — open to data analyst / BI roles
Portfolio Website	your-portfolio.com	Full project portfolio including other Excel & BI projects

Direct Download	github.com/YOUR-USERNAME/.../raw/main/dashboard.xlsx	One-click download of the Excel workbook
Email	your.email@gmail.com	For collaboration, freelance enquiries, or job opportunities

■ Star this project on GitHub if it helped you!

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